CSC 1102: Structured and Object-Oriented Programming Practical Exam Project

Instructions

- You are required to work in your assigned groups to design and develop a simple Java application based on a project idea of your choice.

- Your application should have a simple Graphical User Interface (GUI) using the Java Swing framework and should demonstrate the Object-Oriented Programming concepts covered in this course.

- Your application must satisfy the following requirements:

    - **Object-oriented design:** At least 3 classes that represent meaningful entities in your application; appropriate attributes and methods for the classes; appropriate use of constructors; appropriate use of access modifiers; use of getters and setters where appropriate; creation and use of objects from your classes.
    - **Graphical user interface:** Your application must have a simple GUI developed using Java Swing, which allows a user to interact with the main functionality of your application. For example, a Library Management System might allow the user to: add a book, register a reader, view available books, borrow a book, etc.
    - **Flow control:** Your program should demonstrate the use of conditional and iterative programming constructs covered in class, such as if/else, switch, for, while, do-while. These should be used meaningfully in your application.
    - **Arrays:** Your application should store and manage multiple objects. For example: Book[] books. The application should allow the user to perform operations on the stored objects.

- Please note that:

    - The project must be developed in Java.

- ➢ Do not simply copy an existing project from the internet. You may research how Java/Swing features work, but the application should be designed and implemented by your group.
- ➢ Keep the project simple and achievable. A small application that works correctly is preferable to a large application with incomplete or non-functional features.
- ➢ You are not required to use a database. You may store your data in arrays or other data structures covered in the course.
- ➢ You are not required to implement user authentication, networking, or advanced software engineering features.
- ➢ The GUI does not need to be visually sophisticated. Marks will primarily be awarded for the correct application of Java programming and OOP concepts.

Examples of Project Ideas

You may choose any suitable project. The following are examples only:

- ❖ Library Management System
- ❖ Student Management System
- ❖ Pharmacy Management System
- ❖ Hospital Management System
- ❖ Simple Banking System
- ❖ Inventory Management System
- ❖ Hotel Room Booking System
- ❖ Event Registration System
- ❖ Movie Ticket Booking System
- ❖ Vehicle Rental System
- ❖ Restaurant Ordering System

Submission Requirements

- ❖ Create a GitHub repository (for version control) for your project using one of your group members' accounts. All your project files should be in this repository.
- ❖ Create a short document including a title page with the group details and the GitHub link, a short description of the application, its main features, a list of classes and their

purpose, and how the classes interact. You can include screenshots of your main GUI screens. This should be in PDF format and less than 5 pages (excluding the title page).

- ❖ Submit a zipped file with all your code and the document on MUELE using the submission link that will be provided. Your main file should have the group's details as comments. The name of your zipped file should be the name of your group leader underscore day/afternoon, e.g., kafeero_marvin_day.zip.
- ❖ Each group will be given 15 mins to present their project on a date that will be determined by the lecturer.
- ❖ The deadline is **12th November 2026.**

The purpose of this project is not to build a large or complicated software system. This is a beginner class, and you are being assessed on your ability to apply the Java and object-oriented programming concepts learned in this course to develop a simple working application. Keep your project simple, functional, and well-designed. Good luck!